CHANDERNAGORE COLLEGE

SEMESTER – IV B.Sc. (MINOR) INTERNAL ASSESSMENT EXAMINATION, 2025

SUBJECT : PHYSICS COURSE CODE : PHYS4021 COURSE NAME : THERMAL PHYSICS

FULL MARKS: 10 TIME: 30 MINUTES

ATTEMPT ANY 02 OUT OF THE FOLLOWING 04 QUESTIONS

$2 \times 5 = 10$

Q1 : Show graphically the spectral distribution of Black Body Radiation for at least two temperatures. State Wien Distribution Law for blackbody radiation. Derive the same from Plank's Law. [2+1+2]

Q2 : Which thermodynamic quantity needs to be optimised to have the equilibrium distribution for any statistics? Consider a particular fermionic single particle energy level accommodating 2 identical particles having 3 degenerate levels. Find the number of microstates. Use the Bose-Einstein distribution function for photons and appropriate density of state, to find an expression for the number of bosons between energy E and E+dE, i.e., N(E) dE. [1+2+2]

Q3 : Give kinetic interpretation of temperature on the basis of kinetic theory of gases. Define the mean free path of a gas. Write the expression of mean free path in terms of diameter of gas molecule and particle density. Calculate the r.m.s velocity of an Oxygen molecule at NTP. [1+1+1+2]

Q4. State Zeroth law of thermodynamics. Find an expression for work done by an ideal gas during an adiabatic process. [2+3]

$c = \sqrt{\frac{3RT}{M}}$